

# The Verified-Candidate Loop for Budgeted Large Language Model-Assisted High-Level Synthesis

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**Abstract**—High-level synthesis (HLS) agents must repair and optimize unfamiliar kernels within limited model and tool budgets. The Verified-Candidate Loop (VCL) lets validation evidence control candidate promotion. Its quality-of-results (QoR) score aggregates effective latency, capacity-normalized resource footprint, and source-change evidence into a single bounded utility. QoR-guided retrieval-augmented generation (QoR-RAG) reuses compatible evidence, and Failure Reflection turns tool diagnostics into the next constraint without extra model calls. We evaluate three open-weight large language model endpoints on a balanced 150-task suite spanning six HLS capabilities. We report public-gate completion and a starter-anchored QoR proxy from submission-side synthesis. Qwen3.6-27B achieves the highest completion rate (98.7%, 148/150 tasks). DeepSeek V4 Pro leads the 25-task optimization proxy (85.5) while consuming 5.87M tokens. Qwen3.5-122B-A10B uses the fewest tokens. These single campaigns support endpoint-level comparison; architecture attribution requires controlled repeated trials.

**Index Terms**—high-level synthesis, LLM agent, FPGA, retrieval-augmented generation, quality of results

## I. GATE EVIDENCE, NOT PROMPTS

LLM agents can edit unfamiliar HLS kernels, but only tool reports certify correctness and hardware quality. A metered agent therefore needs a promotion rule that preserves a verified fallback when any required validation gate fails [1].

Prior HLS agents automate code transformation, directive search, and feedback-driven repair [2]–[4]. Tool-using language agents alternate reasoning and actions [5]. Our design assigns candidate promotion to deterministic HLS evidence under explicit model and tool budgets.

The Verified-Candidate Loop (VCL) contributes three findings. **(1) Validation contract.** A fail-closed workflow preserves a verified fallback across seven gates: Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric completeness. **(2) Hardware objective.** The bounded score  $Q_{\text{HW}}$  combines effective latency, capacity-normalized resource footprint, and source-change evidence to reject costly speedups. **(3) Endpoint evaluation.** Qwen3.6-27B achieves the highest public-gate completion rate (98.7%), DeepSeek leads the starter-anchored optimization proxy, and Qwen3.5 uses the fewest tokens (Section IV).

## II. TOOL EVIDENCE CONTROLS THE AGENT WORKFLOW

The task interface exposes an editable kernel, public tests, target part, and budgets while withholding hidden tests and frozen references. The controller records every candidate, gate, metric, token, and credit in one run state.

The VCL runs seven gates in order: Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric completeness. Frequency enforces 100 MHz, Capacity enforces U55C limits, and CoSIM applies only when required. Define validity  $V(c) = \prod_i G_i$  over applicable gates  $G_i \in \{0, 1\}$ ; an inapplicable gate contributes 1. Let  $b$  be the verified fallback. The promotion rule is  $P(c) = V(c) \wedge [Q_{\text{HW}}(c) > Q_{\text{HW}}(b)]$ .

Gate failures fall into three actionable classes, each driving a specific repair strategy without extra LLM calls. **Interface mismatches** (112 of 691 candidates, 16%) trigger API-usage constraints fixing function signatures and pragma syntax. **Synthesis bottlenecks** include timing violations, resource overruns, and II blow-up. QoR-RAG converts them into directive constraints such as pipeline, unroll, and array partitioning. The controller repairs **co-simulation deadlocks** via handshake insertion and port-width correction. Duplicate candidates and repeated failures stop retries.

Figure 1 summarises the loop. Across three campaigns: 691 candidates, 112 interface rejections, 487 synthesis gates, 67 CoSIM gates. Co-simulation failures incur the highest per-repair token cost, consistent with deadlock complexity.

## III. ONE SCORE REJECTS FALSE SPEEDUPS

$Q_{\text{HW}}$  combines performance gain and resource cost into a single bounded utility. Let anchor  $a$  and candidate  $c$  have cycle latency  $L$ , effective period  $p$ , and initiation interval  $II$ . The performance ratio  $r_P = (p_a L_a)/(p_c L_c)$  captures speedup. The area ratio  $r_A$  uses a capacity-normalized footprint  $F(x) = \sum_r x_r / K_r$  over LUT, FF, DSP, BRAM, and URAM to prevent discontinuous jumps from resource transfers. Two evidence factors reward source edits ( $D$ , 1.01 $\times$ ) and validity repair ( $F$ , 2 $\times$ ). The aggregate hardware ratio  $h = 1.01^D 2^F r_P^{0.55} r_A^{0.45}$  weights performance at 55% and area at 45%, calibrated against 99 frozen PPA references.

$Q_{\text{HW}} = 1 - 1/(1+h)^2$  maps  $h=1$  (no change) to 0.75,  $h \rightarrow \infty$  to 1, and  $h \rightarrow 0$  to 0. The mapping degrades continuously for bloated speedups. Budget efficiency  $E = \max(0.80, 1 - 0.10u_{\text{credit}} - 0.10u_{\text{time}})$  discounts excessive tool calls with a floor at 0.80, where each  $u$  is the consumed fraction clipped to  $[0, 1]$ . The final score is

$$S = 100 V(c) Q_{\text{HW}} E. \quad (1)$$

Promotion compares  $Q_{\text{HW}}$  only after  $V(c)=1$ ; a failed gate yields  $S=0$ , and the efficiency floor guarantees a valid baseline at least 60 points.

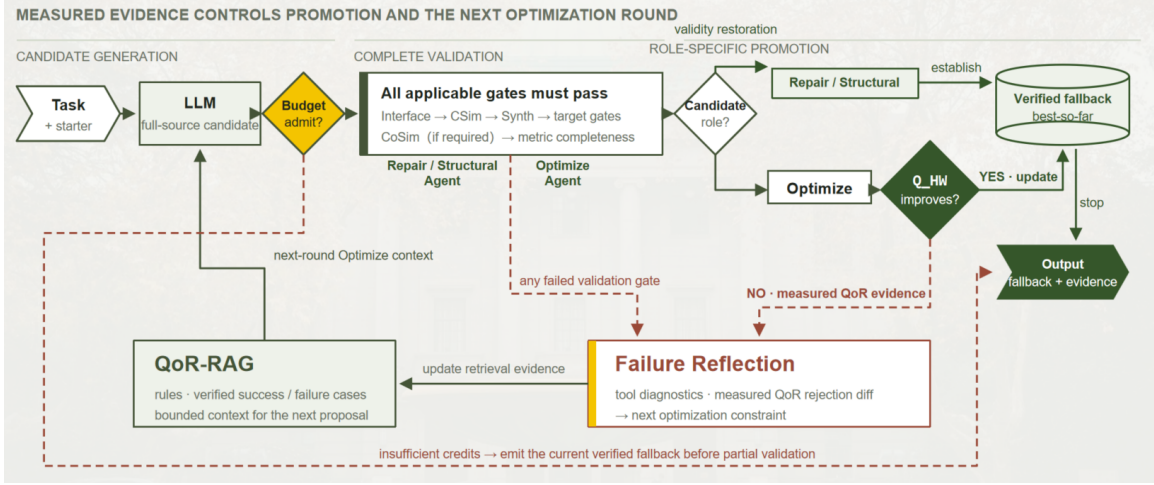


Fig. 1. **The Verified-Candidate Loop.** A candidate replaces the verified fallback only after every applicable gate passes and  $Q_{HW}$  improves.

#### IV. COMPLETION AND PROXY QUALITY FAVOR DIFFERENT ENDPOINTS

**Protocol.** Three hosted endpoints process the same balanced 150-task suite under one agent revision and the Track-A contract [6]. We report public-gate completion and a starter-anchored QoR proxy from submission-side synthesis. The proxy applies Eq. (1) to the 25 explicit optimization tasks; it excludes evaluator-side hidden validation and reference fallback.

TABLE I  
COMPLETION AND PROXY QUALITY FAVOR DIFFERENT ENDPOINTS. THE QoR PROXY IS 100  $Q_{HW}$   $E$ , COMPUTED AGAINST THE STARTER FROM SUBMISSION-SIDE SYNTHESIS FOR THE 25 OPTIMIZATION TASKS.

| Endpoint          | Completed       | Tokens (M) | Credits | QoR proxy | Proxy >76   |
|-------------------|-----------------|------------|---------|-----------|-------------|
| DeepSeek V4 Pro   | 144/150 (96.0%) | 5.87       | 2,617   | 85.5      | 22/25 (88%) |
| Qwen3.5-122B-A10B | 140/150 (93.3%) | 1.68       | 2,375   | 76.3      | 7/25 (28%)  |
| Qwen3.6-27B       | 148/150 (98.7%) | 1.92       | 2,515   | 79.7      | 15/25 (60%) |

**Results.** Table I and the repository’s per-category breakdown yield three patterns.

(1) **Qwen3.6 leads public-gate completion.** Qwen3.6-27B achieves 98.7% completion (148/ 150), including 25/25 on structural CoSim, and leads the other endpoints in code generation. Qwen3.5 records 20/25 structural completions, the lowest category result in the comparison.

(2) **DeepSeek leads the optimization proxy at higher token cost.** DeepSeek reaches mean  $Q_{HW}$ =0.880 and proxy 85.5, with 22/25 (88%) of tasks above 76. Qwen3.6 reaches  $Q_{HW}$ =0.812 and proxy 79.7; Qwen3.5 reaches  $Q_{HW}$ =0.775 and proxy 76.3. DeepSeek uses  $3.1\times$  Qwen3.6’s tokens, while Qwen3.5 uses the fewest tokens.

(3) **The campaigns expose a coverage–quality–cost trade-off.** Qwen3.6 leads completion, DeepSeek leads the optimization proxy, and Qwen3.5 minimizes token use. One campaign per endpoint and different hosted providers prevent attribution of these gaps to model architecture alone.

**Takeaway.** Qwen3.6 leads public-gate completion, DeepSeek leads the starter-anchored optimization proxy, and Qwen3.5 uses the fewest tokens.

#### V. LIMITS AND NEXT STEPS

The VCL makes validation evidence and  $Q_{HW}$  the promotion authority. The capacity-normalized resource footprint replaces worst-single-growth with a continuous aggregate ratio. Qwen3.6-27B leads public-gate completion at 98.7%. DeepSeek V4 Pro leads the starter-anchored optimization proxy at 85.5, while Qwen3.5 uses the fewest tokens. These outcomes separate completion, proxy quality, and model cost.

The proxy omits evaluator-side hidden validation and reference fallback. The agent also depends on mature testbench inputs, a fixed toolchain (Vitis 2025.2), and a single FPGA platform (Alveo U55C). Matched ablations must isolate causal mechanisms: full agent, no QoR-RAG, no measured-failure retrieval, and no Failure Reflection; each must report success, mean  $Q_{HW}$ , tokens, calls, credits, and wall time. Broader corpora and cross-platform transfer remain for future work.

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